

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460676.7435 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.051
Transit depth (Rp/Rs)^2	1.91 +/- 1.42 [%]
Orbital Inclination (inc)	89.89 +/- 2.22
Airmass coefficient 1 (a1)	1.07 +/- 0.026
Airmass coefficient 2 (a2)	-0.06 +/- 0.11
Scatter in the residuals of the lightcurve fit is	2.44 %
Best Comparison Star	#2 - [350, 380]
Optimal Aperture	2.62
Optimal Annulus	11.22
Transit Duration (day)	0.0686 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.051 vs 0.1572 (deviation 0.38σ)
Duration fit vs published	0.0686 d vs 0.075 d (deviation 1.14σ)
Mid-time O–C	11.6 min
Depth SNR	2.7
Residual scatter	2.44 %

Light curve

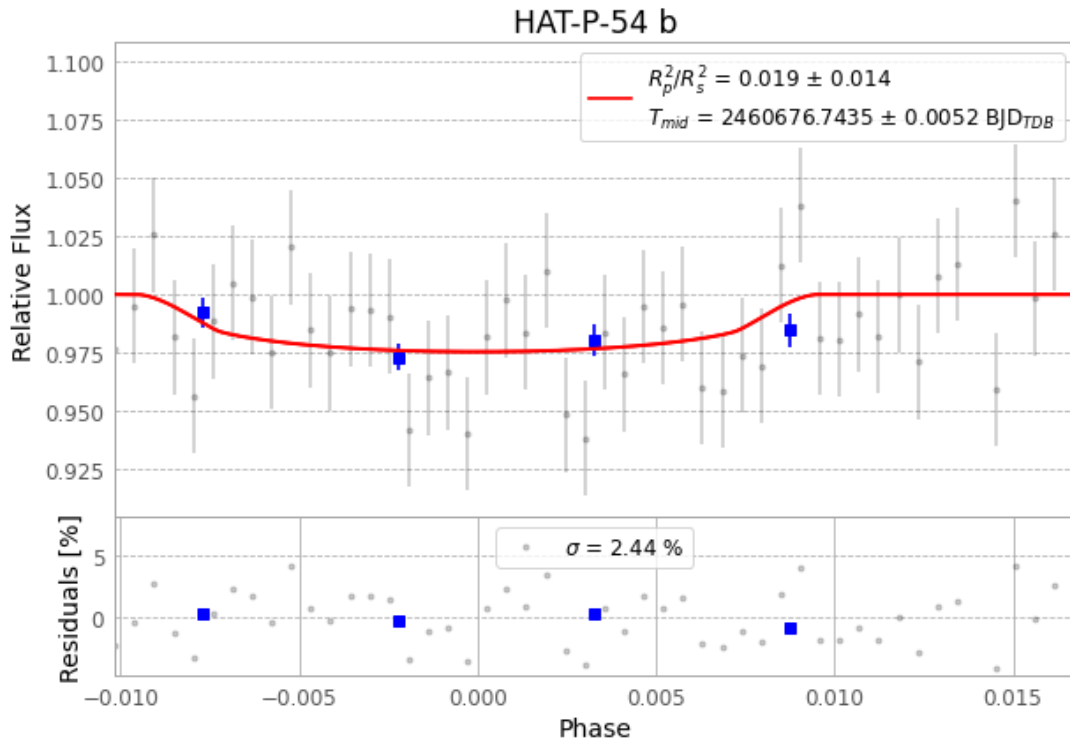


Figure 1 — FinalLightCurve_HAT-P-54 b_2024-12-31.png (SHA-256 bd15a9475e6326e1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [393, 209], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ba40ffe847c508c5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

51 FITS frames (33 MB), HATP-54250101044515.FITS → HATP-54250101071516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-250101.log (0eb0ad6edeed...), FinalLightCurve_HAT-P-54 b_2024-12-31.png (bd15a9475e63...), FinalLightCurve_HAT-P-54 b_2024-12-31.csv (acf0ad8c3916...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 16:36Z.